

Connor Savugot

B.S. in Computer Engineering | North Carolina State University | Graduated May 2026

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TECHNICAL PROFILE

Embedded firmware engineer with experience in multi-MCU firmware boot processes, TI Embedded Linux platforms, programmable power-conversion control, and CAN-connected battery systems. Experienced across C/C++, multi-chip communication solutions, Linux services, board bring-up, bench debugging, and cross-platform Windows/Linux tooling.

TECHNICAL SKILLS

Firmware & OS: Embedded C/C++, Python, Bash; TI Embedded Linux, Yocto/Arago, Linux kernel/device trees, systemd, Docker, FreeRTOS

Processors & Interfaces: TI AM62, TI TMS320 DSP, PIC32, STM32G474RE, MSP430FR2355; CAN/CAN FD/J1939, I2C, SPI, UART/RS-232, ADC, PWM, GPIO, IPMI/IPMB

Development & Test: Git/GitLab, Code Composer Studio, JTAG, KiCad, SocketCAN/PCAN; oscilloscopes, logic/protocol analyzers, through-hole/SMT soldering

ENGINEERING EXPERIENCE

AEGIS Power Systems

Embedded Firmware Engineer

May 2026–Present

- Build and debug TI AM62-based Embedded Linux systems using Yocto/Arago images and toolchains; compile and integrate kernels and device trees, diagnose boot and service failures, and develop systemd services for external-watchdog monitoring and recovery.
- Integrate and debug APU/EPU embedded power subsystems spanning battery hardware, CAN-connected devices, and Linux services; trace faults across power and wiring, bus traffic, Linux configuration, service logs, and application code.
- Diagnose CAN/CAN FD, IPMI/IPMB, I2C, SPI, and UART behavior with logs, register reads, analyzers, scopes, schematics, datasheets, and controlled substitutions to isolate hardware, firmware, timing, and configuration faults.
- Develop C/C++, Python, and Bash software, use Docker in development workflows, and improve code and tooling portability across Windows and Linux.

Embedded Systems Intern

May–Aug 2025

Computer Science Intern

May–Aug 2024

Selected engineering across both internships

- Developed TMS320 DSP firmware controlling transformer-coupled bidirectional buck–boost power stages through 12 PWM A/B pairs (24 physical outputs): independently configured each A output's period/frequency, duty cycle, and relative phase; paired B matched or inverted A.
- Developed multiple connected stages of one PIC32/TMS320 product's firmware-update workflow: a user-facing C++ CLI accepted supplied HEX files, Python/TCP transport provided chunking and checksum validation, and the processors coordinated external SPI-flash/golden-image handling, UART DSP programming, verification, and boot control.
- In 2024, built a restricted SSH/chroot CLI with Python/Bash for TI Yocto/Arago embedded power hardware; reverse-engineered RS-232 transactions and checksums; supported CAN/FreeRTOS debugging, schematic review, soldering, and burn-in-test hardware.

TEAM Industries | Engineering Intern

Jun–Aug 2021 & Jun–Aug 2022

- Developed a repeatable Excel/VBA dimensional and spline-tolerance validation test that automatically flagged out-of-spec parts; built a searchable 17,000+ record tooling/part/vendor catalog with operation, location, and revision traceability.

SENIOR DESIGN

EV Active Sensor Adapter | ST MCU, TI BQ ADC, CAN

NC State University

- Defined architecture and interfaces linking an STMicroelectronics MCU control subsystem, TI BQ-family battery-monitor/ADC, external CAN transceivers, and custom-PCB power, sensing, and telemetry hardware.
- Evaluated component and electrical/protocol tradeoffs; coordinated requirements, data paths, risk tracking, subsystem bring-up and validation, design reviews, and team documentation.

TEACHING & TECHNICAL LEADERSHIP

ECE 306 Lead TA | NC State University

Spring 2025–Spring 2026

- Supported three semesters: two in an unofficial role at about 10 hr/week and one as the official TA at 20 hr/week; delivered test preparation, labs, code/PCB debugging, project guidance, and ECE-program support.
- Taught and debugged vehicle subsystems spanning MSP430FR2355 firmware, ADC line sensors, timer/PWM motor control, GPIO/interrupt inputs, Wi-Fi/serial links, LCDs, and custom PCBs; added supplemental lectures and question-driven support as the class average rose from its typical C+ to A-.

IEEE Open Project Space 2 | Lead Instructor

Fall 2025 & Spring 2026

- Ran two semesters of a project-based electronics course from idea through demonstration; taught firmware, PCB/BOM workflows, and through-hole/SMT assembly while coaching students to isolate and solve their own design problems.

IEEE Soldering Workshops | Coordinator, Lecturer & Lab Instructor

- Coordinated, lectured, and supervised standalone workshops using custom through-hole and surface-mount flashing-light PCB keepsakes; taught professional technique, safety, inspection, debugging, and rework.